DEMONSTRATION CANDIDATE ASSESSMENT BRIEF

Assessment Reference: EDU-RAW-001
Assessment Title: Sine Wave Voltage Divider
Assessment Level: Demonstration / Introductory
Assessment Type: Electronic Circuit Analysis
Analysis Software: EDUPREP Signal Viewer
Filename: EDUPREP_Sine_Voltage_Divider_Healthy.raw

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1. Introduction

You have been provided with a simulated electronic circuit consisting of a sinusoidal signal source and a resistive voltage divider.

Your task is to investigate the circuit's behaviour using the supplied simulation data and the EDUPREP Signal Viewer.

This exercise introduces the basic principles of waveform analysis, electrical measurements and comparison between theoretical calculations and simulated results.

2. Circuit description

The circuit consists of:

- A sinusoidal voltage source.
- Two resistors connected in series.
- A common ground reference.
- An output voltage measured across the lower resistor.

Circuit specifications

Parameter	Value
Input signal DC offset	2 V
Input signal peak amplitude	2 V
Input signal frequency	1 kHz
Resistor R1	10 kΩ
Resistor R2	10 kΩ
Simulation duration	20 ms

The input signal is defined as:

VIN = SINE(2 V, 2 V peak, 1 kHz)

3. Assessment objectives

By completing this exercise, you should demonstrate the ability to:

1. Identify and interpret voltage waveforms.
2. Determine DC voltage levels using waveform averaging.
3. Measure maximum and minimum voltage.
4. Calculate peak-to-peak voltage.
5. Determine signal frequency.
6. Calculate voltage gain or attenuation.
7. Compare measured results with theoretical calculations.
8. Explain any differences between expected and measured results.

4. Assessment tasks

Task 1 – Familiarisation with the EDUPREP Signal Viewer

Load the supplied RAW file.

- Identify the available voltage signals.
- Display the input and output waveforms.
- Identify the input and output measurement nodes.

Task 2 – DC measurements

Use the average measurement function to determine:

- Average input voltage.
- Average output voltage.

Record your results and explain what the average voltage represents in this circuit.

Task 3 – AC measurements

Determine the following for both input and output signals:

- Maximum voltage.
- Minimum voltage.
- Peak-to-peak voltage.
- Peak amplitude.

Record your measurements.

Task 4 – Frequency measurement

Determine the input and output signal frequencies.

Are the frequencies identical? Explain your observation.

Task 5 – Theoretical calculations

Calculate the expected output voltage using the voltage divider equation:

$$V_{OUT} = V_{IN} \times \frac{R_2}{R_1 + R_2}$$

Calculate the expected:

- DC output voltage.
- AC peak output amplitude.
- AC peak-to-peak output voltage.

Task 6 – Comparison and analysis

Compare your calculated values with the values obtained from the RAW file.

Explain whether the simulation results agree with your theoretical calculations.

Task 7 – Investigation

Change the value of R2 from 10 kΩ to 4.7 kΩ.

Before running the simulation:

1. Predict the new output DC voltage.
2. Predict the new output AC peak amplitude.
3. Explain why the output changes.

Run the simulation and verify your predictions using the EDUPREP Signal Viewer:

Filename: EDUPREP_Sine_Voltage_Divider_Faulty_R2_4k7.raw

5. Candidate results sheet

Record your measurements below.

Measurement	Input	Output
DC average		
Maximum voltage		
Minimum voltage		
Peak-to-peak voltage		
Peak amplitude		
Frequency		

6. Candidate conclusion

Provide a brief explanation of:

1. How the voltage divider operates.
2. The relationship between the input and output voltages.
3. How the RAW Viewer measurements compare with theoretical calculations.
4. What happened when R2 was changed to 4.7 kΩ.

7. Submission requirements

Submit your completed measurement sheet and a brief technical explanation of your findings.

Your assessment will consider:

- Correct use of the EDUPREP Signal Viewer.
- Accuracy of measurements.
- Correct theoretical calculations.
- Understanding of circuit operation.
- Ability to interpret and explain simulation results.

End of Assessment Brief